INTERFACE DESIGN PACKET

TAROT

A faithful webOS phone shell for Nura on PinePhone Pro

Cards, gestures, quiet notifications, and the everyday detail that made the original feel complete.

Reference: Pre 2 era webOS 2.x.
Direction: Nocturne.
Canvas: 720 × 1440 physical pixels.



Recreate the phone experience in full

This packet translates the brief into a visual target, an interaction contract, and an implementable handoff.

Tarot should be recognizable as phone-era webOS immediately, and remain recognizable after a day of use. The reference is Pre 2 era webOS 2.x, including stacks. Use the taller display for more content while preserving that structure.

Nocturne is the recommended release theme: deep teal, restrained warm metal, and original botanical linework. It keeps the dimensional light controls and individual icon silhouettes. The Tarot identity appears in the wallpaper, boot mark, and theme palette.

Three kinds of decision

Class	Meaning
Reference	Documented phone behavior and visual anatomy.
Adaptation	A deliberate response to PinePhone hardware or modern clients.
Proposal	A starting value or design treatment that still needs capture or testing.

The packet contains

- 25 editable portrait screen studies and matching 720 × 1440 renders.
- Cards, stacks, Launcher, wave, search, notifications, menus, settings, lock and calls.
- Three theme routes, an original mark, wallpapers, type and material guidance.
- Geometry and motion tokens, service boundaries, failure states, and fidelity acceptance gates.

The screens are design targets. Contacts and app data are fictional. App interiors depict cooperative applications; a compositor cannot automatically restyle every GTK or Qt client.

Start from the actual phone

Use one phone build as the final visual authority. These reference fragments come from Palm's own Pre 2 guide.



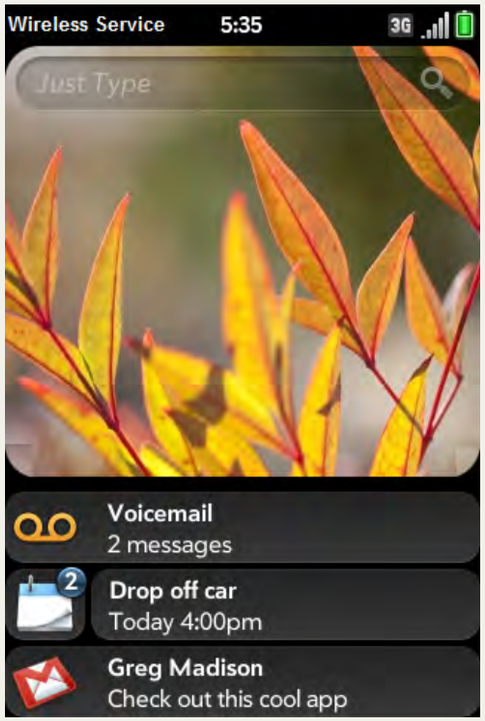
Card view and tray

Printed page 29



Phone Launcher

Printed page 29



Bottom dashboard

Printed page 41



Light preferences

Printed page 199

Read the whole composition: dense black status chrome, tactile light surfaces, freestanding icons, and a bottom notification well. These images are research references; the source bundle contains newly drawn assets.

Later LunaSysMgr code is useful evidence, but its tablet defaults are not a substitute for a phone capture.

The details that carry the identity

Preserve the relationship between surfaces before adding any theme treatment.

1 Black global frame

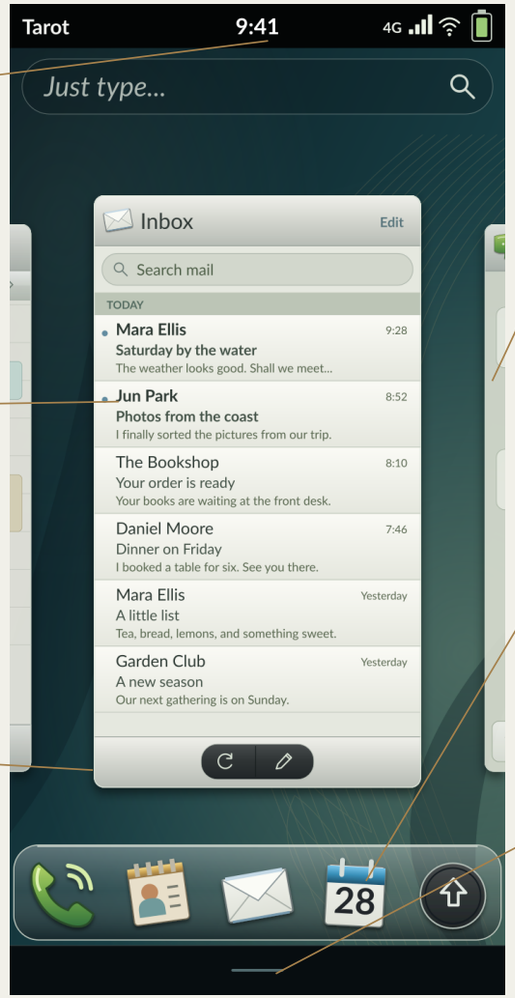
Center the clock. Keep app/menu identity left and device state right.

2 Actual app surfaces

A card is the window itself, scaled continuously. Its contents remain recognizable.

3 Quiet dimensionality

Small corners, soft contact shadows, gentle gradients and thin highlights.



4 The neighboring tasks

Keep lateral peeks and spatial ordering. Focus moves through a horizontal field.

5 Five Quick Launch slots

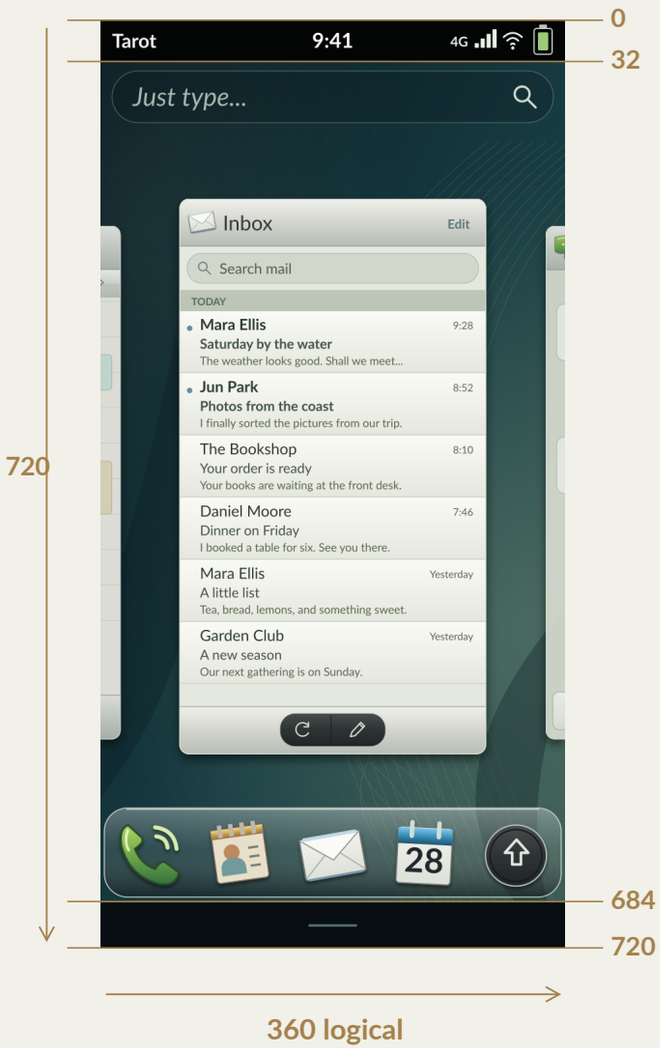
Four favorites plus the Launcher at the right. Give every icon its own silhouette.

6 A reserved gesture strip

A PinePhone adaptation, separated from client input. Keep the center glow quiet.

Scale the interface with intent

All values on this page are proposed Tarot logical units. Output scale 2 produces the native panel raster.

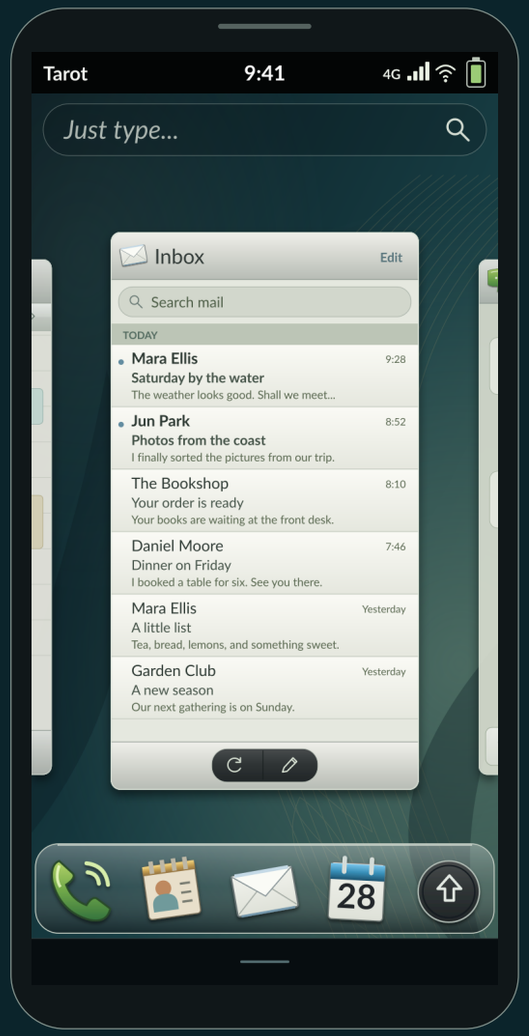


Token	Logical	Physical at 2×	Rule
Output	360 × 720	720 × 1440	Portrait reference
Status bar	32 high	64 high	Black; centered clock
Client rectangle	360 × 652	720 × 1304	Below status, above strip
Gesture strip	36 high	72 high	Exclusive shell input
App header / commands	56 high	112 high	Reference-scale chrome
Focused card	0.66 scale	475 × 861 approx.	Uniform transform
Quick Launch	354 × 69	708 × 138	Five positions; 56-unit art
Banner / dashboard row	32 / 56 high	64 / 112 high	Space grows from bottom
Primary action target	48 min.	96 min.	Compact chrome has alternatives

The historical Pre guide documents 320 × 480, 28-unit bars and 50-unit headers. A width ratio of 360/320 = 1.125 suggests 31.5 and 56.25; Tarot rounds these to 32 and 56. Card geometry and the 36-unit strip remain calibration proposals.

Nocturne in everyday use

The wallpaper supplies the atmosphere. The task surfaces retain their clarity and familiar material.



Card view

One continuous field of tasks.



A focused stack

A related set of windows.



Launcher

Named pages, three columns.

Keep each task spatially continuous

The unit is a task window. A process may own several windows; a transient dialog remains with its parent.



01 In use

One focused window.

02 In Card view

The same surface at 0.66.

03 In a stack

Related windows remain distinct.

Interaction contract

- Tap a card to focus it. Preserve the buffer and the transform origin through the transition.
- Hold, then drag to reorder. Drag onto another task to group; drag away to separate.
- Close one card with an upward flick. Keep stack siblings intact.
- Auto-group only with known launch context. Timing or process identity alone is insufficient.

On Wayland, dismissal requests close. If a save dialog appears or the client refuses, restore the card with its state intact. Force quit is a separate action.

Give the gesture area a precise vocabulary

Shell gestures begin inside the reserved strip. Input that begins in the application remains application input.



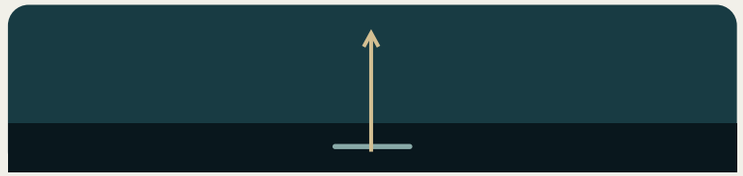
Tap the center

App → Card view; Card view → focused app. Launcher → Card view.



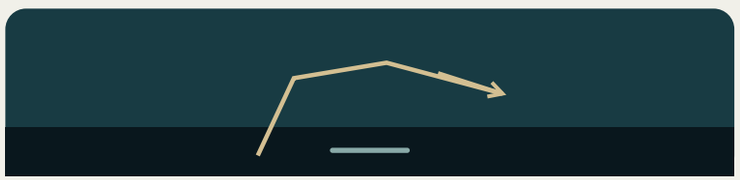
Short stroke left

Back in a cooperating app. At its root, return to cards.



Flick upward

App → cards. Cards → Launcher. Launcher → cards.



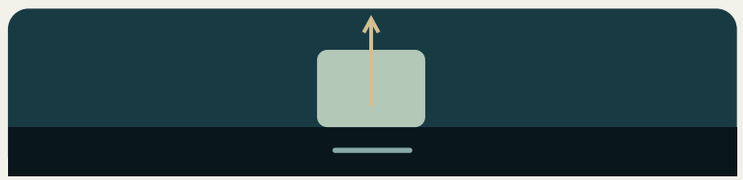
Drag upward slowly

Reveal the wave, slide to a favorite, release to launch.



Full width swipe

Optional direct app switch. Default off; distinguish from short Back.



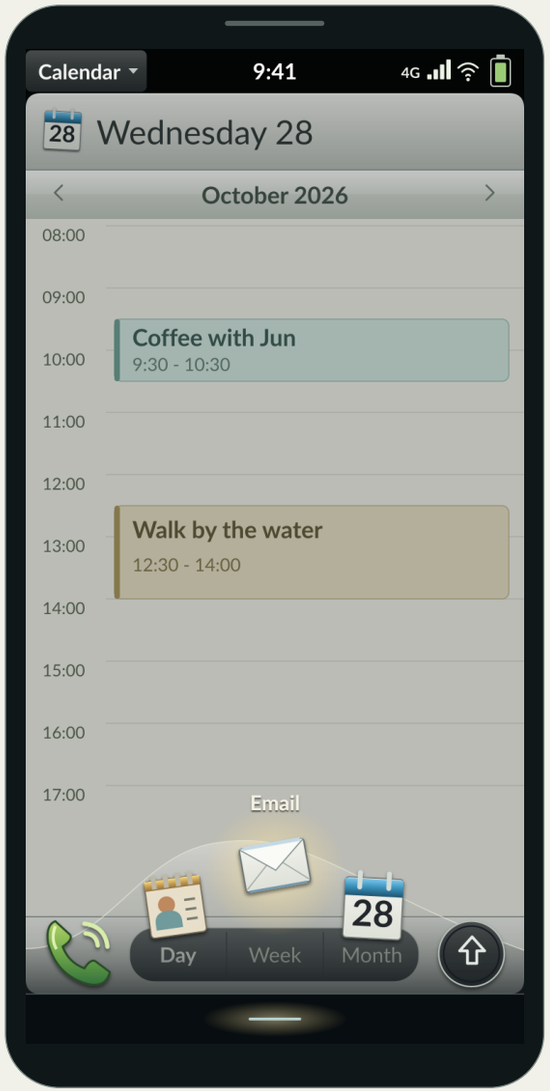
Flick a card up

From Card view only. Close that window after the client accepts.

Initial tuning only: 10-unit tap slop; 14-unit axis lock; 54-unit Back travel; 140 ms wave dwell; 700 units/s home flick; 420 ms reorder hold. Cancel on a second touch and preserve a reversible path until release. These values need real-thumb testing.

The wave is a continuous gesture

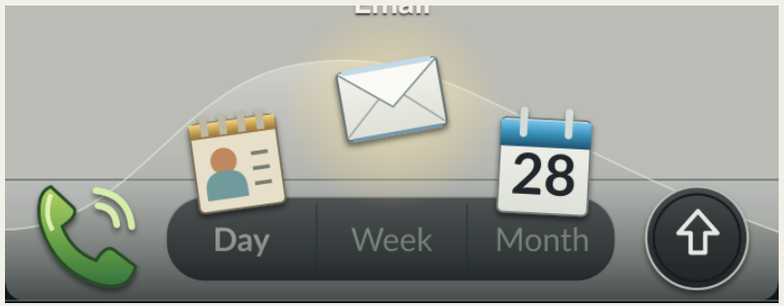
Its shape, icon lift, selection halo and release behavior matter as much as the dock artwork.



Resting tray in Card view



Wave selection above an application



Use the same five favorites and ordering. The selected object rises over the curve. Release while selected to launch; return to the strip and release to cancel.

The resting tray appears in Card view. The ordinary Launcher has its own full page; dragging a favorite can reveal the tray for editing. Expanded notifications temporarily own the bottom well. Wave travel and card motion must remain interruptible.

A page of objects with a clear identity

Preserve the phone Launcher's named pages, three-column rhythm and separate scrolling direction.



Page behavior

- Swipe horizontally between named pages; scroll vertically within a page. A disabled arrow indicates an end.
- Tap an icon to launch. Hold to move it; pause at an edge to move between pages.
- Manage page names and order from the Launcher menu. Keep favorites linked to their original entries.
- There is no persistent Quick Launch dock here. Show it only during favorite editing or an explicit wave gesture.

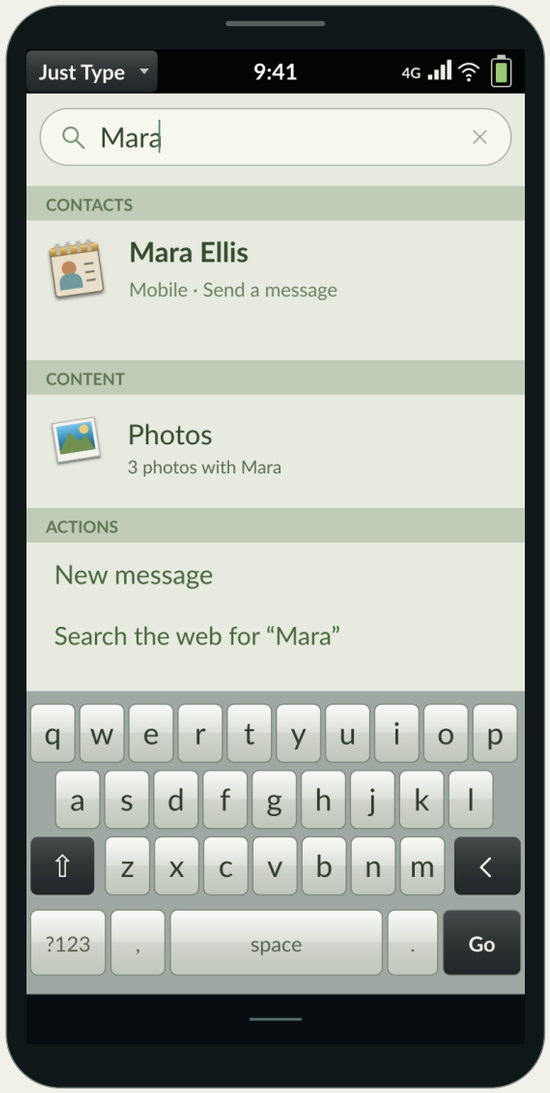
Icon and text rules

Use 58-unit illustrative art in a consistent optical area. Preserve native app silhouettes when there is no Tarot variant. Labels remain readable at 14 units, with up to two lines for long localized names. Crop neither the icon nor its shadow to a universal tile shape.

Launch metadata comes from Desktop Entry files. The page arrangement and favorite order belong to Tarot's local preferences.

Search should lead directly to an action

Typing from Card view or Launcher is part of the reference. Touch keyboard invocation is a Tarot adaptation.



One query, several useful routes

Result class	What selection does
Applications	Launch or focus an application whose name or keywords match.
Contacts / content	Open the exact local record through an enabled provider.
Actions	Create a message, memo or task with the query carried forward.
Search using	Send the text to the chosen provider only after the user selects it.

Keyboard contract

Tap the search field to summon the trusted keyboard. An attached hardware keyboard can begin the query immediately in Card view or Launcher. The keyboard remains above the gesture strip, preserves composition text, and follows focus across app transitions.

The illustrated keyboard is a provisional skin for a compatible input method. Final key geometry, language layouts, candidate rows and text-input protocol versions must be validated together. Local content providers are opt-in; no web query is sent on each keystroke.

Three states in the same bottom space

Arrival, persistence and expansion should feel like one object changing size.



01 Banner

A short message enters below the app.

02 Icon rail

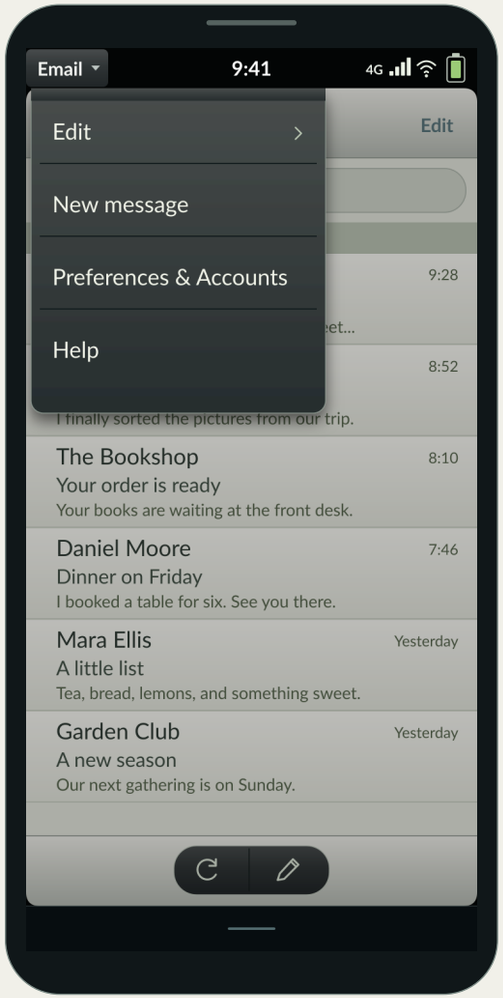
Unhandled activity leaves a compact trace.

03 Dashboard

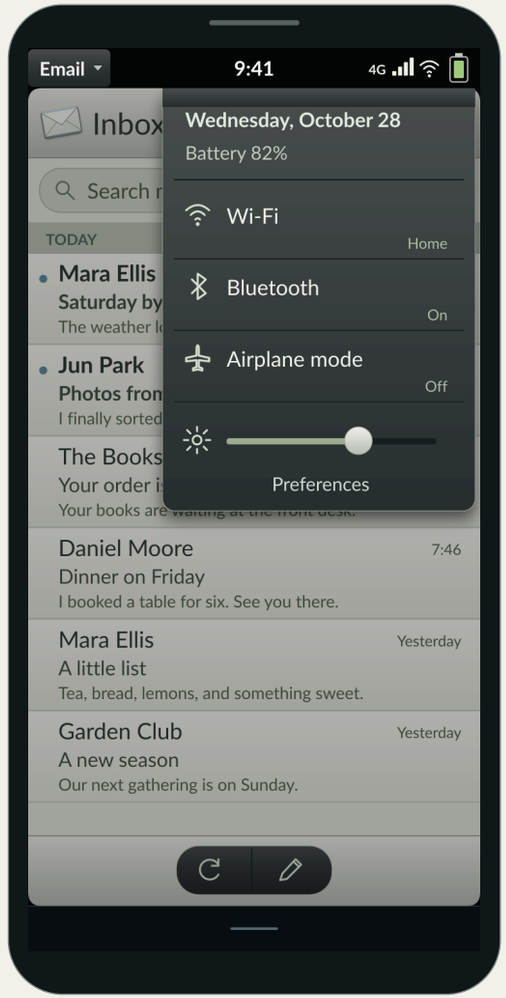
Tap the rail to reveal actionable rows.

Keep the two top corners distinct

The left belongs to the current application. The right belongs to the device.



App commands



Device state

Application menu

Use a compact anchored panel with separators and a real pressed state. Edit leads where supported; scene commands follow, with Preferences or Accounts and Help in consistent positions. The Edit row opens a submenu.

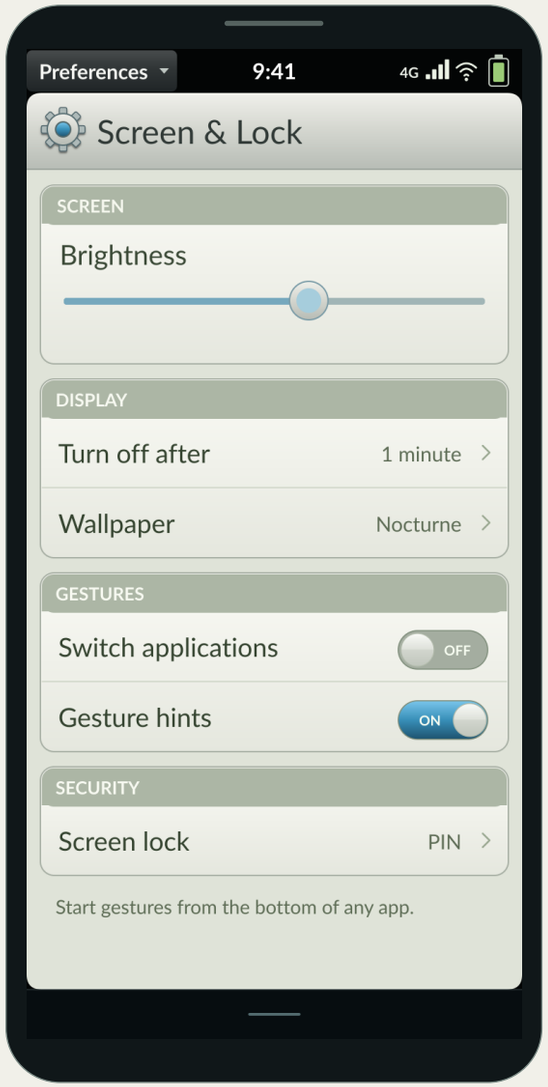
Device menu

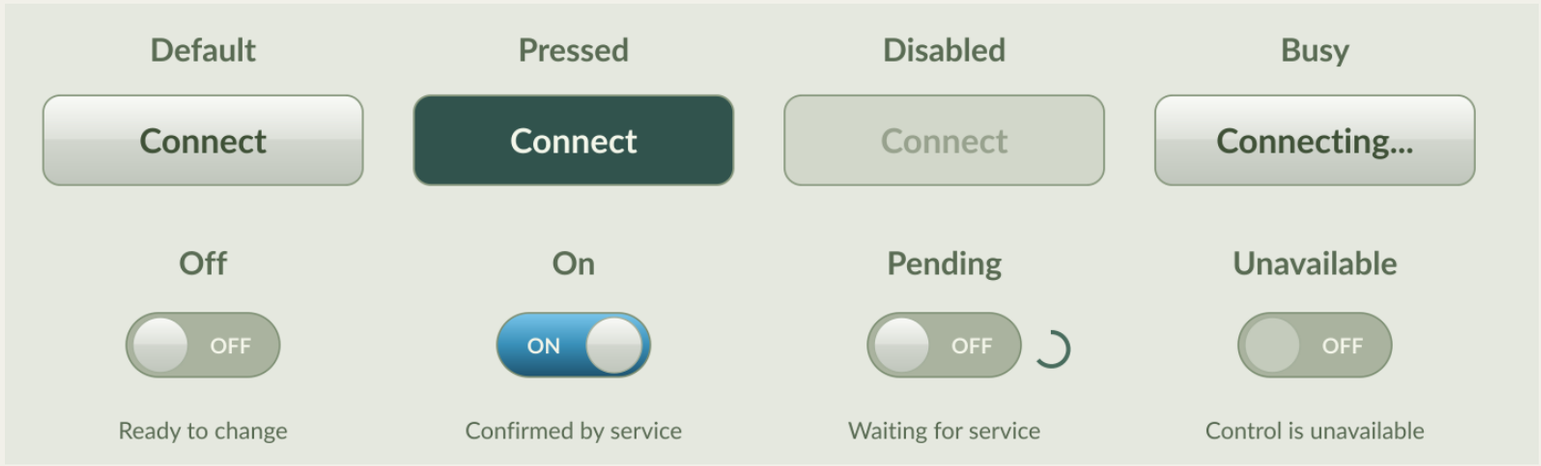
Show date, battery, wireless state, brightness and a route to preferences. Changing a radio shows a pending state until the service reports the result. Expand a row for available networks or devices.

Unadapted clients cannot export arbitrary app commands through core Wayland. Their top-left menu offers genuine shell actions; the client keeps its internal menu.

Make system settings feel native to the shell

Inset groups, small headings, dimensional controls and calm separators are part of the visual vocabulary.





Components need complete states

Control	Required states	Behavior
Toggle	Off / pending / on / unavailable	Observed backend state wins.
Slider	Idle / dragging / focused / blocked	Preview locally; commit safely.
Choice row	Current value / open / selected	Retain the prior value on failure.
Action	Idle / pressed / busy / failed	Use a specific verb and recovery.

Keep the familiar unlock gesture meaningful

Dragging the lock control begins the configured unlock path. It does not bypass authentication.



Presentation

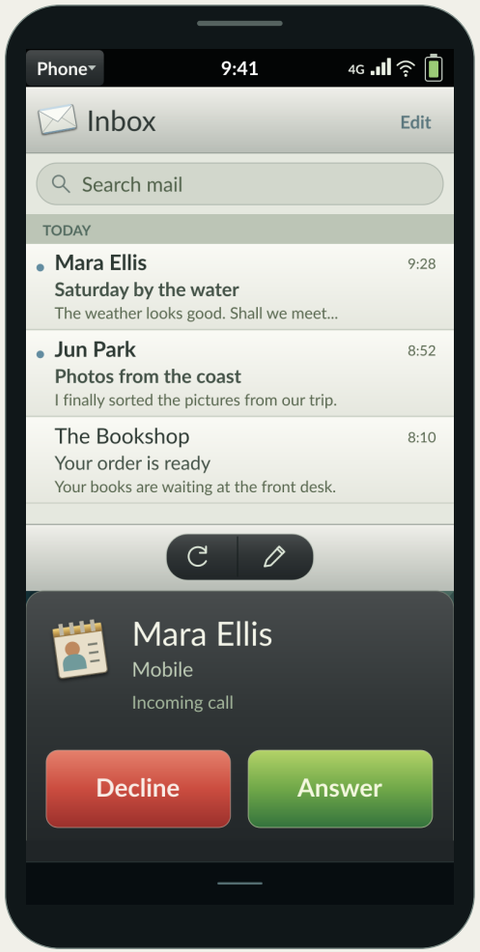
- A quiet clock and date, a movable lock control, and a clear upward target.
- Notification counts are visible; private message bodies stay hidden by default.
- PIN entry provides clear progress and an accessible route to emergency calling when supported.

Compositor contract

Normal client content and input are blocked while locked. Establish a secure frame before revealing a resumed display. A lock-client crash leaves the session locked. The shell must not briefly show a cached card during wake.

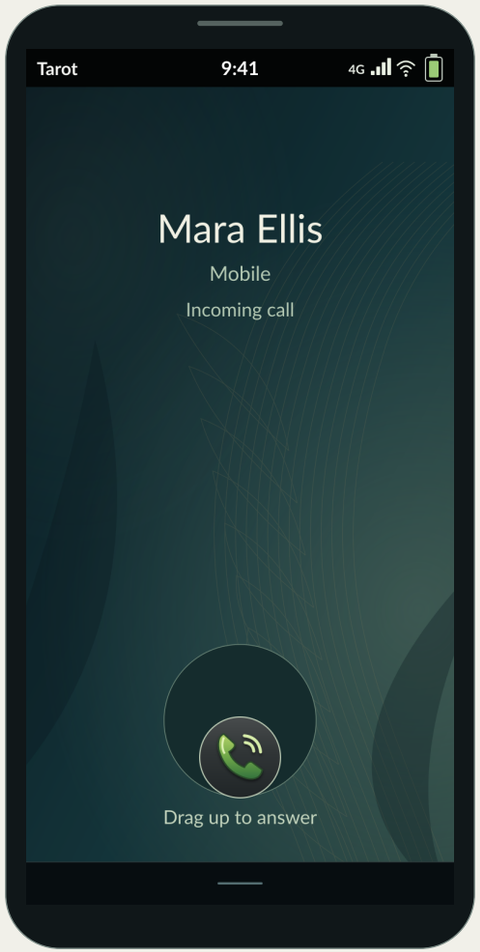
Calls are a first-class interruption

Use the bottom region for an incoming call while unlocked, and a private call surface while locked.



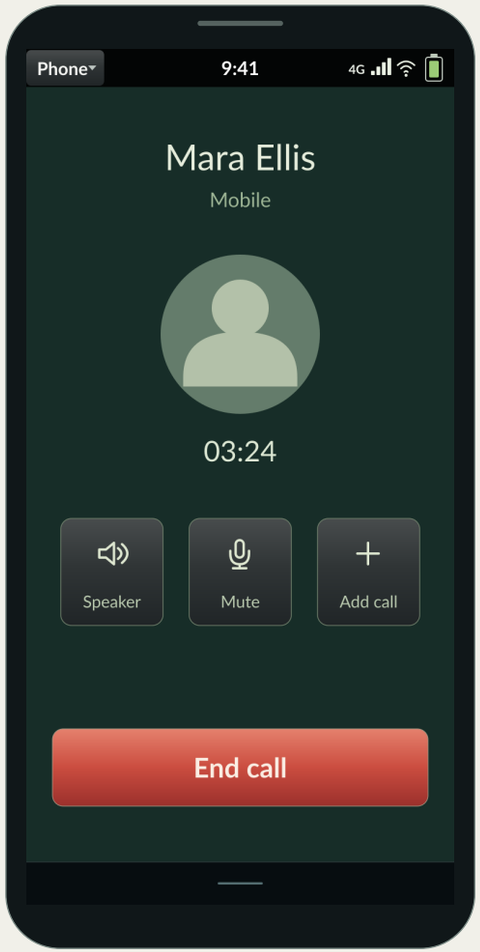
Unlocked arrival

Shrink the active app coherently. Answer and Decline are explicit, reachable actions.



Locked arrival

Drag up to answer without exposing private app content. Provide equivalent accessible actions.



Connected call

Reflect actual call and audio-route state. Mute and speaker changes can be pending or fail.

Be exact about what Tarot can control

The shell can faithfully reproduce navigation and composition. Applications still own the contents of their windows.

Tier	Visual and behavioral promise	Boundary
1 Tarot-owned surfaces	Cards, gesture recognition, Launcher, search, menus, banners, dashboard, lock, call presentation and shell settings.	Full design-system control.
2 Cooperating applications	Matching light chrome, content widgets, app menu, Back semantics and contextual launch into a stack.	A shared style layer and explicit app cooperation are required.
3 Ordinary Wayland clients	Correct card transforms, focus, keyboard, lifecycle and notifications around their existing interface.	Preserve native controls; no invented app menu or universal scene history.

Back must have honest semantics

For a cooperating app, Back traverses its navigation and returns to cards at its root. For an unadapted app, the default fallback returns to Card view. A per-app shortcut mapping is explicit and opt-in; never blindly inject Escape or Alt-Left.

Grouping must have a real cause

Use a known activation or task relationship to place newly opened windows in a stack. Keep dialogs attached to their parent. Do not infer a relationship merely because two windows appeared at similar times.

This packet specifies the style of cooperative Email, Calendar and Messaging screens so the whole experience can be judged. It does not imply that replacing the compositor alone changes those application interiors.

Match the warmth and the metrics

Prelude is the historical target. Lato is the editable substitute used in this packet, and remains a fidelity dependency.

Aa 09:41

Lato Regular / Bold / Italic

The face should feel quiet, human and compact. Compare word widths, numeral rhythm, x-height, punctuation and condensed text against the chosen phone capture. Do not sign off typography by family resemblance alone.

Role	Size	Weight
Status / clock	15 / 16	Bold
Scene title	24	Regular
Main row label	19	Regular
Primary list sender	18	Bold when unread
Secondary / caption	15 / 13	Regular
Launcher label	14	Regular
Card text	Scaled with surface	No separate relayout

TAROT

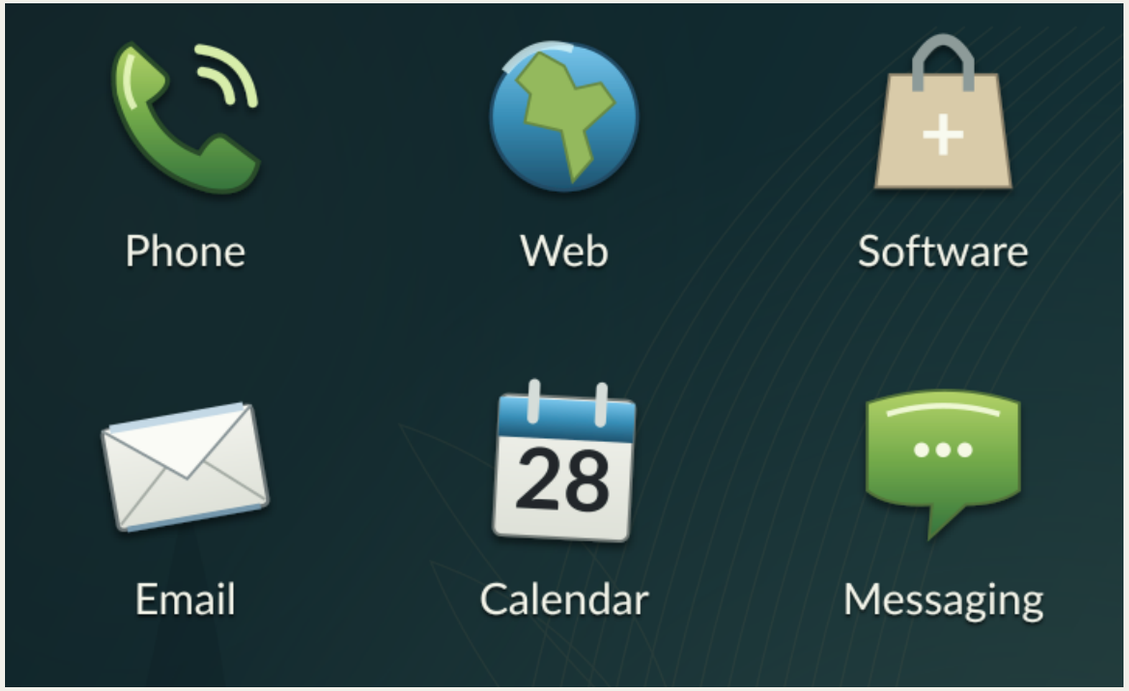
Instrument Serif / identity only

The serif wordmark belongs on boot, About, packaging and wallpaper studies. Keep everyday controls in the UI face.

All sizes are logical units at default scale. Larger text reflows owned panels and rows. Use real text shaping, locale-aware date and time formats, and fallback coverage for the languages supported by the distribution.

Keep the small tactile cues

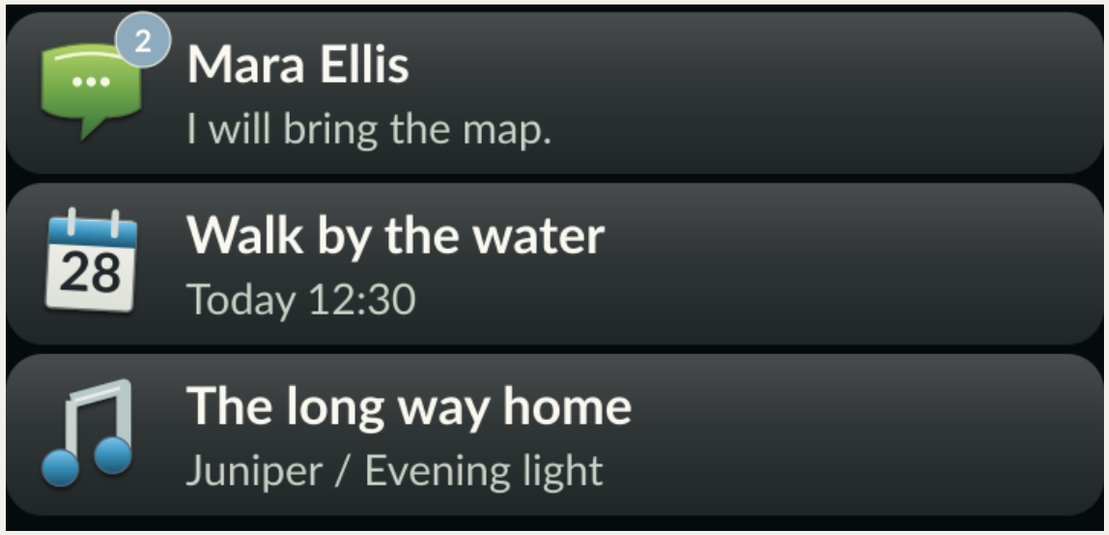
Fine shading, object silhouettes and pressed states give the interface its character at higher resolution.



Application icons

Use distinct objects with light from above, restrained highlights and a small contact shadow. Match optical size, not a common container. Draw native resolutions from vectors; do not enlarge low-resolution historical bitmaps.

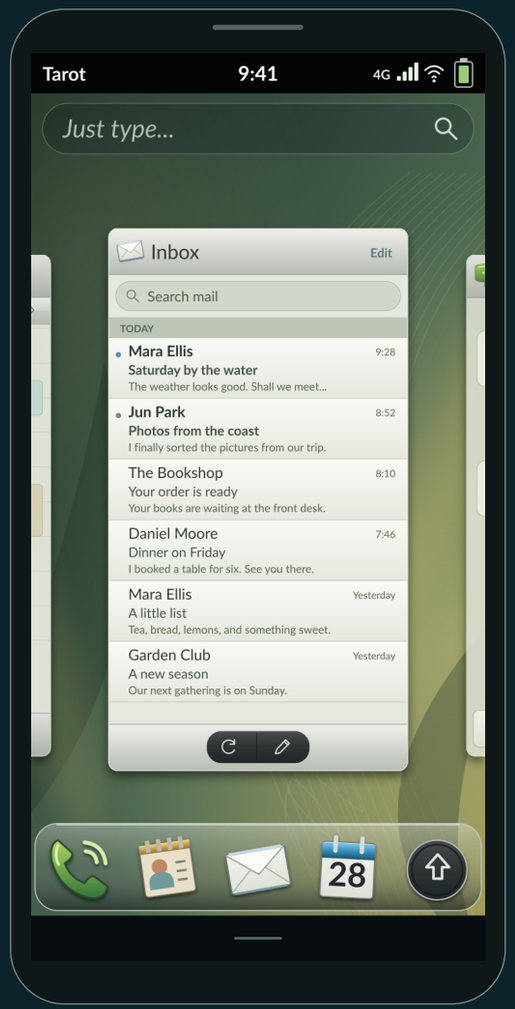




Chrome and dashboard crops shown at large scale for inspection.

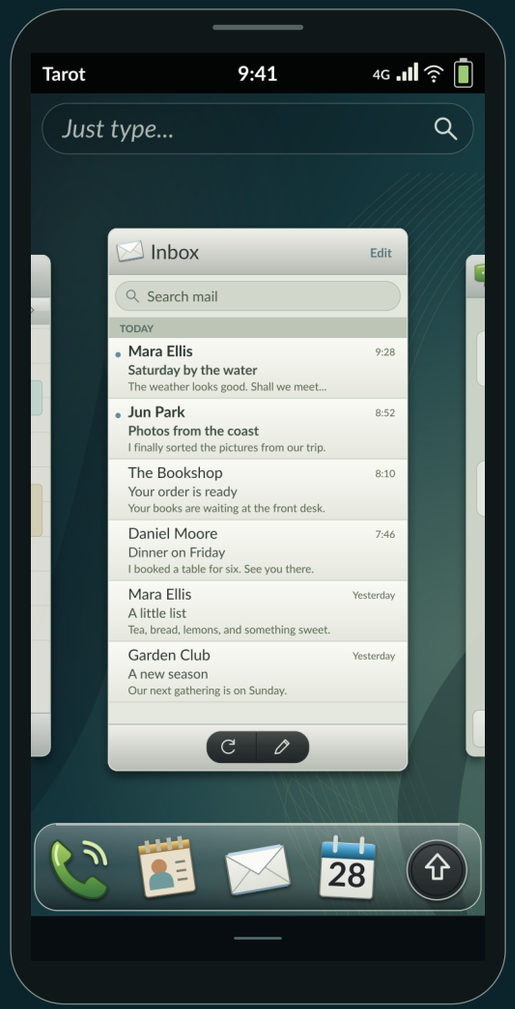
One interface with three atmospheres

Theme changes must not move controls, change gesture semantics or alter the information hierarchy.



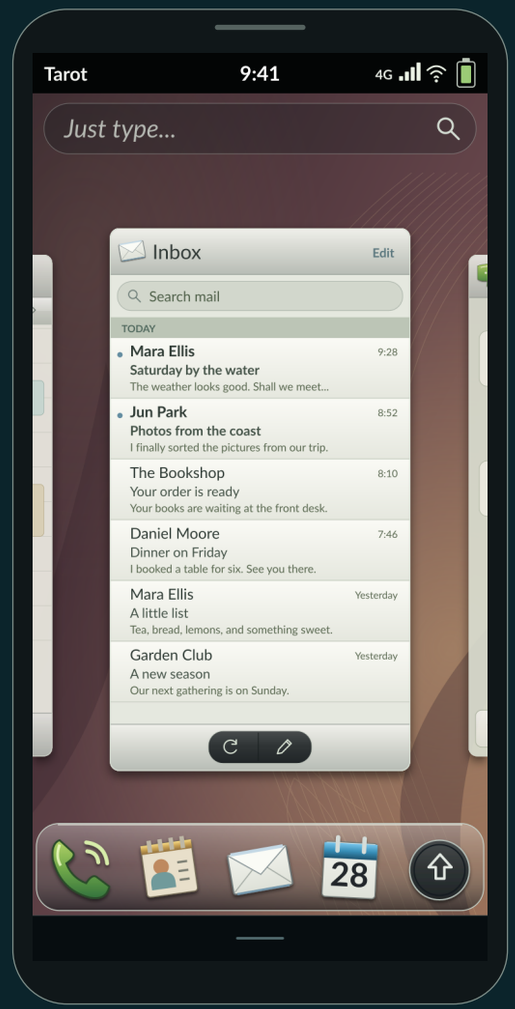
Heritage

Green and soft daylight.



Nocturne

The recommended Tarot default.



Vellum

Oxblood and warm paper.

Let the name live in the material

A restrained card fan is enough to establish Tarot. Ordinary phone functions keep ordinary names.



TAROT

A three-card fan and a small compass-like star.

Themeable surfaces

- Wallpaper and its contrast scrims.
- Accent, focus and selection colors.
- Original app icon variants, boot and About art.
- Bounded material and shadow parameters.

Shared invariants

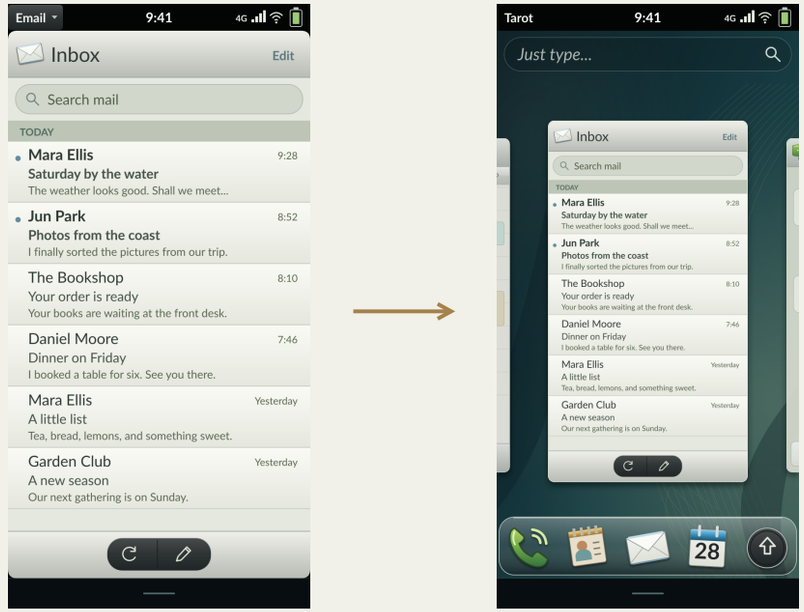
- Status placement and black frame.
- Card scale, gesture meanings and task order.
- Menu anchors and the bottom notification region.
- Legibility, action semantics and lock state.

Animate the task rather than the decoration

Direct manipulation follows the finger. Release transitions settle quickly and keep their spatial explanation.

Transition	Initial duration	Direction
App → Card view	350 ms	Same buffer; uniform reduction
Card → app	300 ms	Same anchor; continuous enlargement
Launch	400 ms	A real new surface enters the field
Dismiss card	300 ms	Follow the upward release velocity
Launcher / Quick Launch	350 ms	Reveal from their spatial origin
Reduced motion	100 ms fade	Preserve state and focus, reduce travel

These duration seeds come from later LunaSysMgr source, not a certified Pre 2 capture. Begin release tuning with a restrained ease-out; preserve live position and velocity when interrupted. Avoid decorative spring overshoot, screen-wide blur and unrelated fades.



Full task **Same task**

The included motion study illustrates the proposed minimize and maximize envelope. It is a design animation, not a recording of Tarot or a measured Palm curve.

Let ordinary services own ordinary services

This is a proposed boundary map. The user has not supplied a repository, so it is not a dependency inventory.

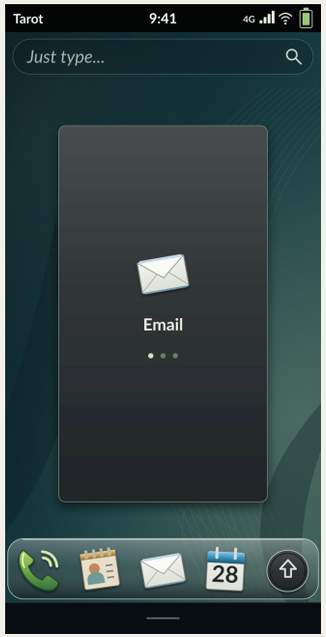
Experience	Standard interface	Tarot owns
Apps and cards	Desktop Entry + Wayland / xdg-shell [R14, R15, R28]	Discovery, launch, focus, transforms, grouping and close requests.
Wi-Fi and data	NetworkManager [R16]	Selection, secrets prompts, progress and truthful state.
Calls and SMS	ModemManager [R17]	Call / message presentation and supported actions.
Audio	PipeWire + WirePlumber [R18]	Volume, routes and call controls through distro policy.
Bluetooth	BlueZ [R19]	Pairing, discovery and connection feedback.
Power and session	UPower + login1-compatible backend [R20, R21]	Battery, inhibitors, suspend and wake presentation.
Notifications	org.freedesktop.Notifications [R22]	One session server; banners, rail and dashboard.
Music dashboard	MPRIS [R23]	Render metadata and supported playback actions.
Text and permissions	Wayland input method + portals + polkit [R24, R26, R30]	Trusted input surfaces and coherent authorized prompts.

Keep service calls asynchronous. Nura can package UIs for OpenRC, systemd, or both; decide what Tarot supports explicitly. Do not add a new modem or audio policy daemon merely to reproduce the old presentation.

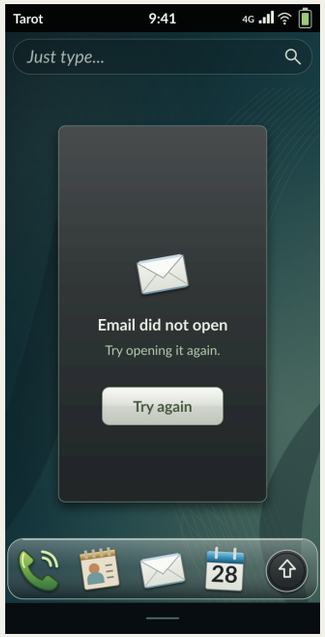
A beautiful shell must tell the truth

Transitions from a request to a confirmed service state are part of the interface, including failure and reconnection.

Situation	Visible response	Recovery
Radio changed	Pending row, then observed on/off state	Revert with a useful explanation if denied.
Service disappears	Unavailable or reconnecting	Retain intent; refresh from new state.
App launch is slow	A loading card using the app identity	Show first real buffer as soon as ready.
App fails to launch	App-specific failure in the card	Try again or dismiss; keep other cards.
Close is refused	Return the card and its dialog	Let the app save or cancel the close.
Low memory	Honest retained snapshot or unavailable preview	Never pretend a dead app is live.



Waiting



Recovery

The wallpaper and other cards stay responsive while a service is busy. Report a hardware switch as the cause only if the backend can actually distinguish it.

Notifications retain replacement IDs and explicit actions. Preserve active non-expired items in the rail; clear dismissed items correctly. A history feature can be separate, but must not silently turn an expired notification into a live request.

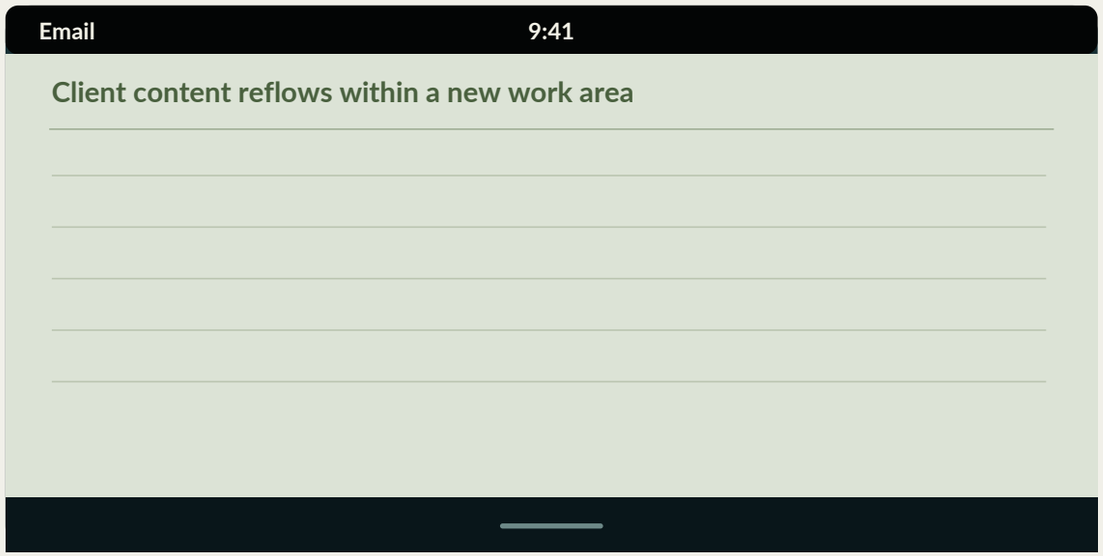
Preserve the grammar while making room

The default studies prioritize the familiar composition. Accessible settings enlarge and simplify without changing navigation.

Text and targets

- Primary actions use 48-unit targets. Compact status chrome, gesture strip and keyboard keys need equivalent accessible routes.
- Larger text grows rows and panels. Wrap names and values; do not solve overflow by shrinking essential text.
- Target 4.5:1 for small text and 3:1 for meaningful graphics. Measure the least favorable gradient and wallpaper region.
- Offer reduced motion, visible gesture hints, keyboard navigation and a screen-reader-ready semantic shell tree.

Landscape is an explicit layout



720 × 360 logical; gesture strip stays at the visual bottom

Portrait remains the reference orientation. On rotation, cancel an in-progress shell gesture, relayout at a stable state, and preserve the focused task. Treat minimum-size desktop apps explicitly; do not stretch their buffers to fill.

The landscape diagram is a layout rule, not a finished keyboard or application study.

Spend the budget on contact and continuity

The high-resolution treatment should improve edge quality while the shell stays responsive and frugal.

Area	Implementation intent	How to judge it
Frame pacing	Render to the actual display cadence; do not inherit the old source FPS limit.	At 60 Hz the frame interval is 16.67 ms. Record missed frames and p95 frame time.
Card composition	Reuse client buffers, clip corners, cache shadows, and damage only what changed.	Inspect drag, stack movement and incoming banners together.
Resizing	Animate a stable surface while negotiating the target size; reconcile new commits.	No black flash, accidental stretching, duplicate input or torn chrome.
Idle power	Static wallpaper and chrome; no perpetual shimmer or repeated full-screen blur.	Compare redraw count and power at idle, overview and lock.
Input feedback	Show contact immediately; do not block drawing on D-Bus or app startup.	Measure touch-to-visible response on the device, not just desktop emulation.
Secure surfaces	Treat lock and privileged input as real compositor state.	Lock during every transition; test crash, wake and focus changes.

These are acceptance targets, not measured Tarot results. Validate the chosen Nura image, kernel, firmware and services on the PinePhone Pro. Calls, wake, sensors and audio routing must be exercised together; a successful desktop mockup does not certify them.

Finish the shell in visible slices

A small coherent surface should become exact before a wide collection of screens becomes approximate.

Slice	Build	Definition of done
01 Reference and frame	Pin the reference phone build; capture its states; implement scale, status bar, client bounds and gesture strip.	Native screenshots agree on geometry; input never leaks across the strip boundary.
02 The card experience	Launch, minimize, maximize, scroll, group, unstack, reorder and request close.	One buffer remains one task through every transition; refused close is recoverable.
03 Everyday shell	Launcher, Quick Launch wave, Just Type, app/device menus and bottom notifications.	The complete daily navigation loop works with keyboard and service interruption.
04 Phone and trust	Lock, authentication, incoming/active/missed calls, audio routes, pairing and permissions.	Locked wake is private; real calls and service failures remain comprehensible.
05 Conformance and themes	Font matching, icon refinement, motion capture, accessibility, Nocturne and alternate palettes.	A recorded audit passes; theme changes preserve the same geometry and behaviors.

Keep a dedicated plain reference theme while tuning geometry. Apply Nocturne after that baseline is stable, then compare both against the same test sequence. This keeps theming from disguising an interaction mismatch.

Define what nearly indistinguishable means

The goal needs an evidence trail: a named reference build, matched state captures and repeatable interaction videos.

Still images

- Match status baselines, icon spacing and menu anchors.
- Compare card radius and shadow falloff at 1× and at native output.
- Check Launcher pages, tray visibility and the wave silhouette.
- Check banner, icon rail, dashboard and urgency states.
- Test long names, empty lists, disabled controls and loading cards.

Motion and input

- Record minimize, maximize, throw and aborted throw.
- Record stack focus, reorder, merge and unstack.
- Separate slow wave, fast flick, Back and advanced switch.
- Interrupt each animation without jumping or losing focus.
- Verify notification space changes while keyboard and calls are active.

System behavior

- Exercise app startup failure and unsaved-document closure.
- Restart services; confirm honest pending and unavailable state.
- Lock during card motion; wake without leaking app content.
- Test calls after suspend, audio routes and modem return.
- Verify large text, reduced motion and external keyboard navigation.

Packet status: researched and visually reviewed design proposal. Not yet certified against one captured Pre 2 firmware build or implemented on the user's device. The remaining visual dependencies are font metrics, icon micro-detail, measured card/stack geometry and measured motion curves.

Historical and visual evidence

Primary authored documents and upstream source. Titles are clickable; full URLs are in the source manifest.

R1 **Palm Pre 2 User Guide**

Palm, 2010; manufacturer PDF hosted by Rogers. Printed pp. 22-42, 52-56, 69-75, 199-200.

R2 **User Interface Summary**

HP developer documentation, 2011; preserved mirror. Reference dimensions and scene anatomy.

R3 **Menus UI Guidelines**

HP developer documentation, 2011; preserved mirror. Menu ordering and anchors.

R4 **Dashboards and Notifications**

HP developer documentation, 2011; preserved mirror. Bottom surfaces and urgency.

R5 **Just Type**

HP developer documentation, 2011; preserved mirror. Actions, content and search providers.

R6 **Human Interface Guidelines**

HP developer documentation, 2011; preserved mirror. Non-blocking notifications.

R7 **LunaSysMgr Settings.cpp**

Open webOS source. Font names and card-scale seeds; later than Pre 2.

R8 **LunaSysMgr StatusBar.cpp**

Open webOS source. Phone black fill and clock placement; later than Pre 2.

R9 **LunaSysMgr animation configuration**

Open webOS source. Provisional duration seeds, not phone motion certification.

R10 **Palm Pre User Guide**

Manufacturer guide hosted by Bell. Earlier phone gesture vocabulary.

R11 **LunaSysMgr repository**

Official Open webOS source and shell responsibilities.

Historical illustrations on page 3 are attributed research excerpts from the Palm guide. The editable Tarot artwork is newly drawn. Later open-source defaults are labeled as seeds and are never presented as measured Pre 2 values.

Platform and asset evidence

The service map is a proposed integration contract based on the interface owners' documentation.

R12 **PinePhone Pro specifications**
PINE64.

R13 **Project rebrand Nura**
Nura, 27 Sep 2026.

R14 **Wayland architecture**
Wayland project.

R15 **Desktop Entry specification**
freedesktop.org.

R16 **NetworkManager developer resources**
NetworkManager.

R17 **ModemManager D-Bus reference**
ModemManager.

R18 **WirePlumber documentation**
PipeWire project.

R19 **BlueZ Adapter API**
BlueZ upstream.

R20 **UPower Device interface**
UPower.

R21 **login1 D-Bus interface**
systemd project.

R22 **Desktop Notifications specification**
freedesktop.org.

R23 **MPRIS specification**
freedesktop.org.

R24 **XDG Desktop Portal documentation**
Desktop portals project.

R25 **ext-session-lock-v1**
Protocol author documentation.

R26 **Smithay input-method implementation**
Smithay.

R27 **Nura service-manager support policy**
Nura, 10 Jul 2026.

R28 **xdg_toplevel close request**
Smithay generated protocol bindings.

R29 **Google Fonts source families**
Lato and Instrument Serif.

R30 **polkit overview**
polkit project.

R31 **callaudiod upstream status**
Upstream README marks it deprecated; new call-profile policy is in WirePlumber..

R32 **Wayland protocol reference**
Wayland project.

A concrete starting point for Tarot

The package is designed to be inspected, changed and implemented without reconstructing the design from screenshots.

Deliverable	Contents	Status
Design packet	31 pages of reference, art direction, interaction specification and handoff.	Reviewed design proposal
screens/svg	25 editable portrait studies at 360 × 720 logical units.	Original vector art; live text
screens/png	Matching 720 × 1440 screen renders.	Ready for comparison and issue discussion
tokens/tarot.tokens.json	Geometry, type, palette, theme and motion seeds.	Provisional implementation values
brand + wallpapers	Card-fan mark, wordmark and three wallpaper themes.	Editable SVG plus wallpaper PNGs
motion	A minimize/maximize motion study at the proposed timings.	Illustrative; not measured hardware footage
handoff + fonts + source	Engineering notes, capture audit, source manifest, font notices and generators.	Reusable implementation reference

The next fidelity decision

Pin the exact webOS 2.x phone build to emulate, then compare this packet's geometry and motion against controlled captures. Keep Nocturne as the lead theme while settling Prelude-compatible typography, the app cooperation surface and real-device gesture tuning.